

Class 8

The American Invention Machine

Part II — Building the American Growth Machine

The American Economy — draft course materials (proposal under discussion)

The point of today's class

America's innovation advantage came from an institutional ecosystem — patents, universities, immigration, federal science, corporate labs, venture capital, and open markets — not from a single inventor or a single law.

Why it matters

- ▶ Vote: which invention would you *least* want to live without?
- ▶ The real question is not who invented it — it is: **what kind of society repeatedly produces, attracts, funds, and spreads such inventions?**
- ▶ Class 3 called ideas “recipes” in the abstract. Today the recipe machine gets concrete — and American.
- ▶ AI is stress-testing this machine in real time.

Today

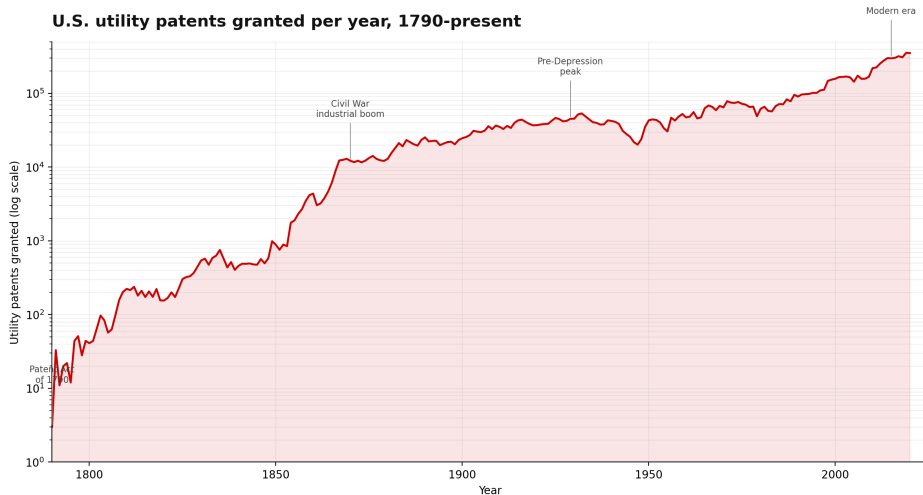
1. Why ideas are economically special
2. Patents: the bargain
3. Who gets to invent?
4. Science becomes an American institution
5. Immigrants and invention
6. GPTs, AI, and the policy toolbox

Ideas are recipes — and markets underproduce them

Good	Rival?	Easy to exclude?
Sandwich	yes	yes
Clean air	no	hard
Formula for a drug	no	sometimes
Search algorithm	no	partly

- ▶ **Nonrivalry** (Romer): a machine helps one worker; an idea can help *every* worker at once.
- ▶ **Spillovers**: the inventor bears the cost; society gets much of the benefit — so markets underproduce ideas.
- ▶ That gap is why innovation policy exists.

U.S. patents granted per year, 1790–present



Source: U.S. Patent and Trademark Office, *U.S. Patent Activity 1790–present*.

The patent bargain

Benefit	Cost
Rewards inventors	Temporary monopoly
Encourages disclosure	Higher prices
Attracts investment	Litigation and patent trolls
Clarifies ownership	Can block follow-on innovation

- ▶ A patent is *not* innovation — it is a legal tool for encouraging and organizing it.
- ▶ Wright Brothers: patents claim particular technical solutions; early aviation litigation may have slowed follow-on flight.
- ▶ Moser (World's Fairs): patent laws shape the **direction** of invention, not the level.

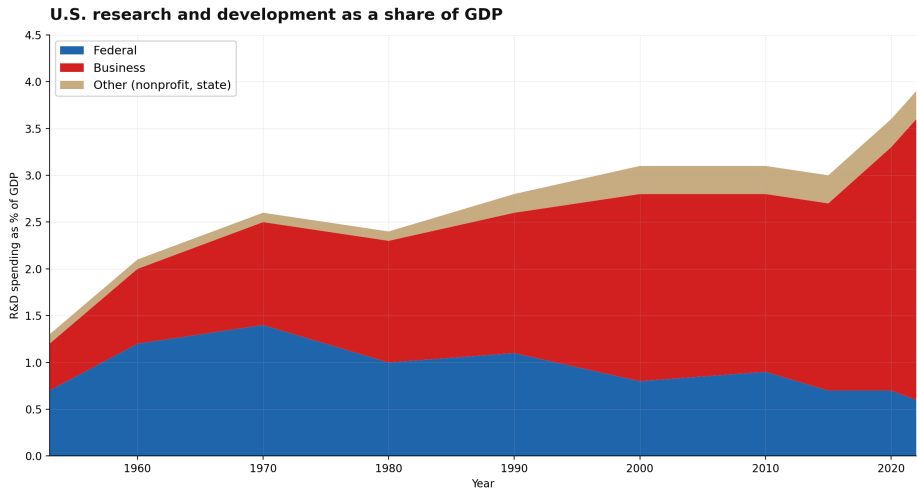
Who gets to invent?

- ▶ “Lost Einsteins” (Bell, Chetty, Jaravel, Petkova, Van Reenen): a child's chance of becoming an inventor varies sharply with parental income, race, and gender.
- ▶ Children *exposed* to inventors in a field are more likely to patent in that same field.
- ▶ **Talent is everywhere; exposure and opportunity are not.**
- ▶ 1870–1940 (Sarada, Andrews, Ziebarth): inventors were disproportionately mobile and foreign-born — by 1940 roughly $1.65\times$ as likely to be foreign-born.

Science becomes an American institution

- ▶ Vannevar Bush, *Science: The Endless Frontier* (1945): scientific progress is a proper concern of government.
- ▶ WWII mobilization — radar, penicillin, the Manhattan Project — then NSF, NIH, DARPA, and the research university.
- ▶ 2023 U.S. R&D: \$937B total — \$138B basic, \$174B applied, \$625B development; the business-funded share has risen sharply.
- ▶ Not laissez-faire, not central planning: a layered ecosystem of markets, universities, firms, philanthropy, and public funding.

U.S. research and development as a share of GDP



Source: National Science Foundation, National Center for Science and Engineering Statistics (NSF 25-334).

Immigrants and invention

- ▶ Moser, Voena, Waldinger: Jewish scientists expelled from Nazi Germany raised U.S. patenting roughly 31% in emigre fields.
- ▶ Immigration is not just labor supply — it shifts the *direction* of science, builds networks, and attracts more talent.
- ▶ Modern echo: the foreign-born share of Nobel laureates, startup founders, STEM graduate students, and AI researchers.

General-purpose technologies take time

GPT	Core invention	Complementary investments
Steam	engine	coal, factories, railways, mechanics
Electricity	generation, motors	grids, factory redesign, appliances
Automobiles	internal combustion	roads, oil, suburbs, insurance
Computers	chips, software	skills, firms, networks, databases
Internet	packet switching, web	broadband, platforms, logistics
AI	models, chips, data	data centers, energy, workflow redesign

- ▶ The invention is only the beginning — productivity comes when the economy *reorganizes* around it (Moser–Nicholas on electricity).

Harder ideas, bigger bets: AI as the live test

- ▶ Bloom, Jones, Van Reenen, Webb: research effort keeps rising while research productivity falls — sustaining Moore's Law now takes more than $18\times$ the researchers of the early 1970s.
- ▶ Stanford AI Index: \$285.9B of U.S. private AI investment in 2025 — more than $23\times$ China's; 1,953 newly funded U.S. AI companies.
- ▶ Is AI more like the light bulb, the internet, the airplane, or the Segway?
- ▶ Distinguish: invention, adoption, complementary investment, measured productivity, welfare.

The innovation-policy toolbox

Tool	Strength	Weakness
Patents	decentralized incentives	monopoly, litigation
Prizes	clear target	hard to set the prize
R&D subsidies	lower the cost of research	may fund what firms would do anyway
Universities	people and open science	commercialization gap
DARPA model	high-risk missions	hard to replicate
Immigration	expands the idea frontier	political backlash

- There is no single best innovation policy: different tools solve different incentive and information problems.

Discussion

Congress wants to double American innovation over the next generation. Where should the next dollar go — patents, universities, immigration, R&D subsidies, prizes, DARPA-style agencies, venture capital, K–12 STEM, or data centers? Defend your answer with one course concept: spillovers, incentives, human capital, market size, nonrivalry, or opportunity.

Read before next class

- ▶ Bell, Chetty, Jaravel, Petkova, and Van Reenen (2019), “Who Becomes an Inventor in America?” (skim the figures).
- ▶ Moser, Voena, and Waldinger (2014), “German Jewish Emigres and U.S. Invention” (skim the introduction).